Week 49: Four towers and differences

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-49-facilitator.pdf, whose source is editable/facilitator-src/content.json with route-note.json. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and the termination proof are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Run four starts to zero	P1	P1	P1	first
Orders of 0, 1, 2 and 4	P2			first
Which order lasts longest?		P2		first
A start that lasts long	P3	P3	P2	first
Starts that become zero in one move	P4			first
Starts that first become zero on the second move	P5			first
Which rings can be the result of one move?		P4	P3	return
Four stations and three	P6	P5	P4	return
Can the tallest tower grow or stay?		P6		return
Even and odd rings			P5	return
Doubling every height			P6	return
Can a height exceed the largest start?			P7	return
Does every start finish?			P8	return

First route. K–1 P1–P5; Grades 2–3 P1–P3; Grades 4–5 P1–P2. **Return visits.** K–1 P6; Grades 2–3 P4–P6; Grades 4–5 P3–P8.

Key, one entry per problem

Run four starts to zero

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. The four starts reach all zeros in 3, 2, 2 and 4 moves; running the rule by hand makes its one-old-ring discipline concrete. Grades 4-5 run only (1,1,0,0) and (0,0,0,2), 3 and 4 moves, to reach their own problems sooner.

Orders of 0, 1, 2 and 4

K–1 P2 · first route

Idea. Of the 24 orders of 0, 1, 2 and 4 around the stations, eight last 7 moves and sixteen last 4: the order, not just the heights, decides the length.

Which order lasts longest?

Grades 2–3 P2 · first route

Idea. Of the 24 orders of 0, 1, 2 and 4 around the stations, eight last 7 moves and sixteen last 4: the order, not just the heights, decides the length.

A start that lasts long

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P2 · first route · wording differs by band

Idea. With heights 0 to 4 (or 0 to 5) the longest runs last 7 moves; searching for long runs is the experiment behind the termination theorem.

Starts that become zero in one move

K–1 P4 · first route

Idea. A start becomes all zeros in one move exactly when all four heights are equal; working backward is an inverse problem young children can solve with towers.

Starts that first become zero on the second move

K–1 P5 · first route

Idea. A start first reaches all zeros on move two when its gaps are all equal but its heights are not, such as alternating heights.

Which rings can be the result of one move?

Grades 2–3 P4 · Grades 4–5 P3 · return route

Idea. (2,0,2,0) and (1,1,1,1) have old rings; (1,1,1,0) cannot, since the gaps around a ring have an even total, and (0,0,0,2) cannot, since three zero gaps force a constant ring.

Four stations and three

K–1 P6 · Grades 2–3 P5 · Grades 4–5 P4 · return route · wording differs by band

Idea. (1,0,0,0) on four stations reaches zero in 4 moves, but (1,0,0) on three stations cycles forever: the number of stations changes what can happen.

Can the tallest tower grow or stay?

Grades 2–3 P6 · return route

Idea. A gap is never larger than the larger of its two heights, so no new tower is taller than every old one; the tallest height can stay for a move, as (0,1,2,4) becomes (1,1,2,4).

Even and odd rings

Grades 4–5 P5 · return route

Idea. Under the even/odd rule every one of the 16 letter rings becomes all E within four moves, so after four moves every height is even.

Doubling every height

Grades 4–5 P6 · return route

Idea. Multiplying every height by k multiplies every gap by k, so a scaled start takes exactly as many moves.

Can a height exceed the largest start?

Grades 4–5 P7 · return route

Idea. A gap between two heights from 0 to M is at most M, so the largest height never grows however many moves are made.

Does every start finish?

Grades 4–5 P8 · return route

Idea. Every four-station start reaches all zeros: four moves make every height even, halving gives a start with the same moves, and the bounded maximum must eventually halve to 0.